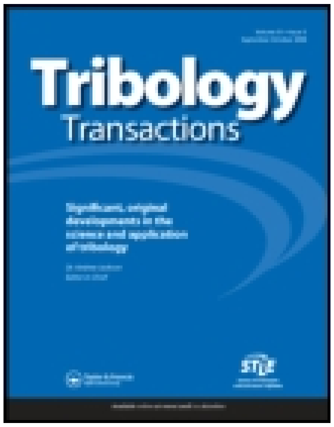

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Starvation and Reflow in a Grease-Lubricated Elastohydrodynamic Contact

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Starvation and Reflow in a Grease-Lubricated Elastohydrodynamic Contact©

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The lubrication mechanisms of a grease in a rolling-element bearing has been studied through the measurement of film thickness in a rolling point contact. To simulate bearing conditions the contact runs under fully starved conditions; there is no attempt to maintain bulk flow of the grease into the inlet using an external supply. In consequence the film thickness drops off rapidly as the contact progressively starves. After a few minutes rolling (at constant speed) an equilibrium film thickness is attained which has two components: a residual film (h_R) comprised of degraded grease thickener and a hydrodynamic component (h_{EHD}) due to the liquid phase from the grease. The hydrodynamic contribution represents a balance between lubricant lost from the contact and replenishment from the grease close to the track. The ability of the grease to replenish the rolling track has been inferred from measurements of lubricant reflow around the static contact. These results are discussed in light of current starvation and grease lubrication models.

KEY WORDS

Elastohydrodynamic Lubrication, Starvation

INTRODUCTION

The current understanding of the mechanism of grease lubrication is woefully inadequate. It is still not possible to establish the nature of the lubricating film, identify the film formation and supply mechanism, nor predict bearing lubrication performance from simple grease characterization tests. The lack of understanding of the way in which a grease lubricates successfully means that the mechanisms, and prevention, of failure are also inexplicable. This is widely recognized to represent a design limitation for many systems. In this respect, the relationship between grease rheology, lubrication mechanism and performance is crucial to the understanding of this problem and it is these areas that this paper begins to address.

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There is little consensus in the literature as to the mechanism of film formation and replenishment in grease-lubricated bearings. Three mechanisms commonly suggested are

1. The separating film is comprised of both thickener and base oil components.
2. The grease acts as a reservoir to provide a controlled supply of oil which bleeds into the contact zone (*1*).
3. Separation is due to a thin layer of highly worked grease deposited in the rolling track during the first few hours of running (*2*).

Experimental evidence in support of these theories is unsatisfactory. Scarlett (*2*), in one of the few direct studies of grease movement in bearings, reported that neither the entire grease nor base oil migrates from the housing recesses to the track. From his own experimental observations, Scarlett concludes that the third mechanism operates; that lubrication is due to a thin layer of highly worked grease deposited in the rolling track during the first few hours of running. It is not thought necessary for the film to be constantly replenished through oil movement from the pack as the grease that has cleared into the housing recesses forms a tightly packed seal which prevents oil leakage from the rolling track.

It is unlikely that lubrication in a bearing is due to flow of the entire grease through the contact; film thicknesses and heat dissipation would indeed be too great if this was the case. However, in many experimental measurements of grease EHD films, the grease is constantly replenished within the rolling track, thus ensuring that the inlet of the contact is fully flooded (*3*), (*4*). This is clearly a poor simulation of grease behavior within a bearing where there is usually no mechanism for bulk replenishment. If an external supply is not used, grease is rapidly pushed to either side of the rolling track and the ensuing inlet starvation gives a significantly decreased film thickness (*5*)–(*7*). This is usually below the limit of conventional interferometry or other measurement techniques (~80 nm) and has proved difficult to study experimentally.

EHD film thicknesses measured with a supplied inlet are usually far greater than those from the base oil alone (*5*), (*6*). Films are typically in the range 0.150–2 μm which corresponds to a specific film (lambda) value of 5-65 for a bearing contact which is clearly not representative of the actual lubrication condition. In tests where the supply has been maintained, infrared spectroscopic analysis of the inlet lubricant has shown that both thickener and base oil are present (*8*). Therefore, it is to be expected that it is the combination of base oil and thickener properties and their resistance to shear that determines the effective inlet viscosity and, hence, film thickness in these tests.

Studies of grease lubrication under starved conditions (inlet supply not maintained) have generally been shown film thickness to decrease with time as the bulk grease is progressively pushed out of the contact and the inlet contracts (*6*), (*7*), (*9*), (*10*). Rasteger and Winer (*5*) measured starved steady-state EHD film thicknesses for a series of greases. They found that starved film thicknesses were significantly less than fully flooded ones ranging from 140 nm for calcium, clay base and aluminium complex greases down to 80 nm for polyurea and lithium greases. As 80 nm represents the lower limit of the technique, the actual film thicknesses might well be significantly less. More recent work by the author has shown that typical film thicknesses of 20–40 nm are obtained after prolonged running under starved conditions (*11*). One observation of this work was that grease films within the contact consist of two phases: a residual solid layer that maintains separation even in the static contact and a classical hydrodynamic component presumably from the base oil. Analysis of residual films within the rolling track has shown that significant amounts of thickener are present, presumably deposited by the grease as it is shear degraded in the contact (*11*).

Most of the film thickness measurements in EHD tests have been at bulk ambient or close to ambient temperatures. This is not representative of conditions in an operating bearing where steady-state running temperatures of 50°–70°C are to be expected. The tests reported in this paper have therefore been run at three temperatures—20°, 50° and 80°C—as these are typical of the bulk temperatures experienced by a grease in its operating cycle from start-up to steady-state running.

The mechanism of film replenishment in grease lubrication has also been investigated in this paper. Replenishment is usually considered to be due to spontaneous bleeding of base oil from the grease replenishing the rolling track and contact. There is little experimental evidence to support this theory and, although there is undoubtedly a replenishment mechanism, it is not due to spontaneous flow of oil from the grease. Rather, an external force (overrolling action of the surfaces, local temperature gradients or surface forces) is required to extract the oil. Åström and coworkers (*6*) in their study of starvation effects in an EHD contact discussed possible replenishment mechanisms. They suggested that primary reservoirs, which form on either side of the contact, govern replenishment and that surface tension effects close to the contact form the local supply mechanism.

Film thickness measurements were therefore carried out without an external mechanism to resupply the rolling track with grease. This is considered to be representative of the steady-state lubrication condition of a bearing although it is, perhaps, the worst case. In a real bearing, operating and design parameters might combine to provide intermittent grease redistribution and contact supply. Such mechanisms are not considered in this paper.

From the above it can be seen that there are two important areas of interest:

1. The nature and composition of the separating film
2. Mode of replenishment

These areas are addressed in this paper. The aims of this study therefore, were two-fold to measure lubricant film thickness under starved conditions at realistic bearing operating temperatures and to correlate these results with those of lubricant reflow around the contact.

EXPERIMENTAL

Test Conditions and Materials

Lubricant film thickness has been measured for a rolling point contact at three bulk temperatures: 20°, 50° and 80°C. A diagram of the test device is shown in Fig. 1. Thin film optical interferometry was used to measure film thickness in the center of the contact. This method has been described in earlier papers (*12*). It is capable of measuring films in the range 2–300 nm with a resolution of 2 nm. The bearing contact is simulated by a steel ball loaded and driven in nominal pure rolling by a rotating glass disc. The underside of the disc is coated with a chromium semireflective film overcoated by silica. The experimental test conditions are summarized in Table 1. The grease tested was a commercial lithium hydroxystearate (~12% w/w) with an NLGI grade of 2–3. Its

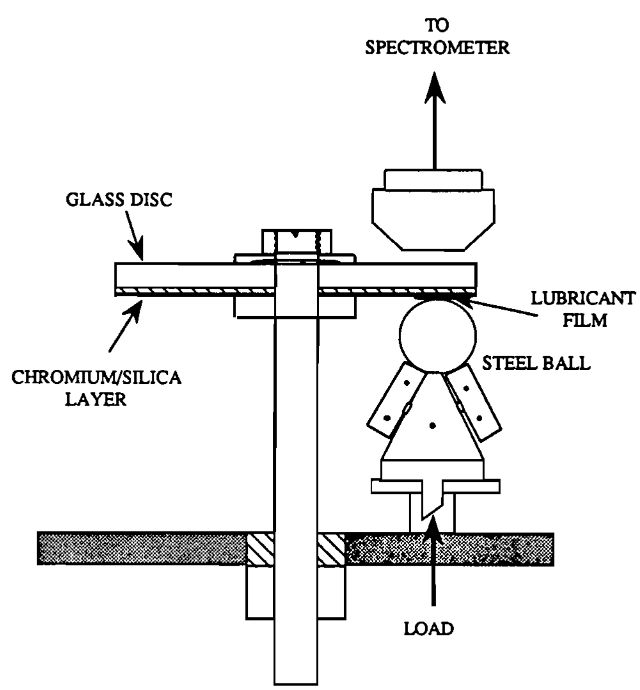

Fig. 1—Film thickness measurement in a rolling point contact.

TABLE 1—EHD TEST CONDITIONS	
Bulk Temperature	20°, 50°, 80°C
Rolling Speed	0.025, 0.05, 0.1 m/s
Duration	30 minutes
Max Hertz Pressure	0.48 GPa

base oil is a hydrorefined mineral oil with a viscosity of 46 cSt at 40°C.

Film Thickness Measurement

In this paper, film thickness measurements have been made for a grease under fully starved conditions. There has been no attempt to resupply grease to the inlet or rolled track. Unless the inlet is constantly replenished, it progressively starves and the film thickness drops. Measured film thickness under these conditions is both time and rolling speed dependent. To obtain a characteristic film thickness for a particular rolling speed it is necessary, therefore, to run the test at constant rolling speed until an equilibrium film thickness has been reached. Observation of grease behavior in these tests suggests that overrolling of the ball squeezes the grease back toward the track. Closer to the contact, surface and capillary forces combine to form a local supply around the contact. In extended tests the grease closest to the contact is continuously worked by the overrolling action of the ball. This results in a breakdown of the grease structure and, hence, improved release of oil for contact replenishment. The effect of this on film thickness has therefore been investigated through extended tests at different bulk temperatures representative of those found in operating bearings.

The test procedure therefore was as follows. The steel ball and glass disc were cleaned by solvent washing, air dried and then assembled. The test device was heated to the required temperature and allowed to stabilize. A layer of grease was applied to the track while the disc was rotated slowly by hand. The temperature of the grease was allowed to equilibrate—this was monitored by a thermocouple in the inlet region—before the test was started. The motor was switched on and set to the required speed. Film thickness measurements were taken every minute over a period of 30 minutes. Three values were taken and averaged for each measurement. At the end of the test, the motor was stopped and the residual film thickness at zero speed determined.

EHD film thickness was also measured for the grease base oil at the three test temperatures. The EHD contribution from the base oil alone can now be estimated for each of the grease test conditions. This is an important parameter as the base oil film thickness is often taken as an indication of lubrication levels for bearing applications.

Earlier results (*8*) have shown that grease lubricating films are two-phase: a solid deposited layer (h_R) that is probably composed of degraded thickener particles and an EHD component (h_{EHD}) due to the base oil in the track or supplied from the grease reservoir. The composite film thickness can be represented as

$$h_T = h_R + h_{EHD}$$

where h_T is the *total* film thickness measured during rolling in the EHD tests.

It is possible to estimate h_R by taking a film thickness measurement at zero speed and this has been done at the end of the tests. However, it must be remembered that this measurement is taken at only one point on the ball and can only give a general indication of the residual film thickness. h_{EHD} is the difference between the film thickness measured during rolling and the residual film measured statically. This film (h_{EHD}) is therefore generated by the rolling action of the ball and is considered essentially hydrodynamic in origin.

Lubricant Replenishment

It is more difficult to quantify lubricant replenishment to the contact. If local capillary flow around the contact is a replenishment mechanism, it is almost impossible to measure under dynamic conditions. It can only be quantified indirectly by measuring resulting changes in film thickness. Observation of grease behavior in a starved EHD test has shown that once rolling has stopped, oil bleeds out of the grease close to the contact reforming a meniscus within the rolled track. If this also occurs continuously for the dynamic condition, it might be considered to be a local lubricant replenishment mechanism. It might be possible to infer any changes in this supply condition by measuring the time taken for the meniscus to reform when motion has been halted.

Lubricant replenishment was quantified in a separate series of tests by stopping the rig and observing oil reflow around the static contact. The time taken for the meniscus to reform with oil bleeding from the surrounding grease was measured, see Fig. 2. A microscope with graticule eyepiece was used to quantify reflow. The time taken for the meniscus to reach an arbitrary distance from the center of the contact was recorded. The tests were run in the same way as the film thickness measurements except that the motor was stopped every few minutes to measure reflow time.

RESULTS AND DISCUSSION

EHD Film Thickness Results

EHD film thickness results are shown for both the base oil and grease in Figs. 3–6. Film thickness/rolling speed log-log

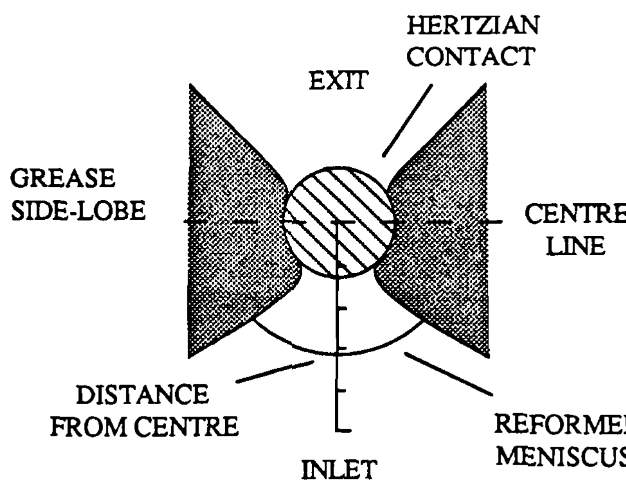

Fig. 2—Measurement of inlet reformation.

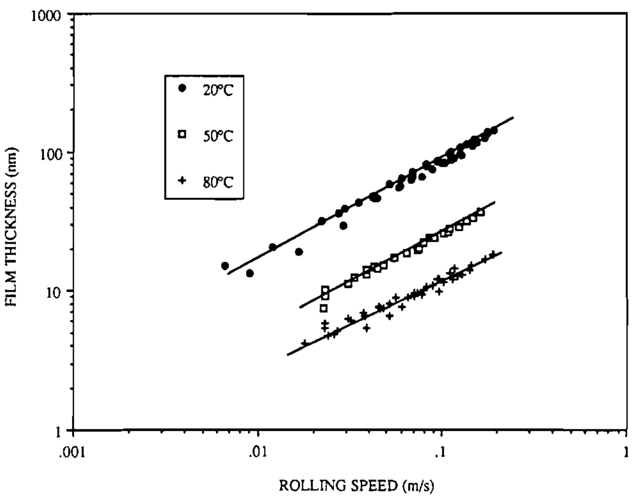

Fig. 3—EHD film thickness results for base oil.

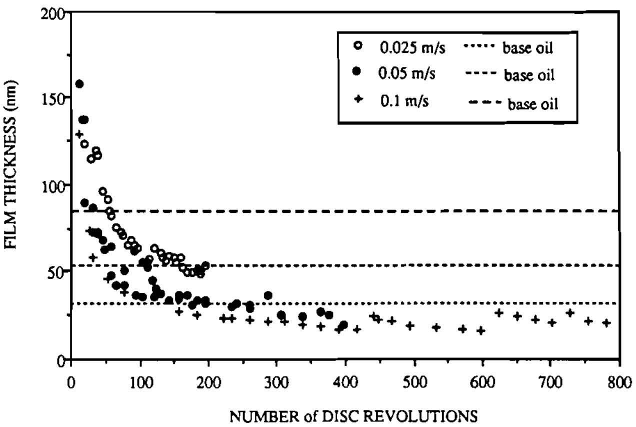

Fig. 4—Film thickness results for grease and base oil at 20°C.

plots are shown for the base oil at different temperatures in Fig. 3. There is no evidence of starvation and the curves are representative of the fully flooded condition. As expected, film thickness rises with increasing speed and drops rapidly with increasing bulk temperature.

Representative film thickness measurements for the grease at three different temperatures are shown in Figs. 4–6. Note that film thickness is plotted against number of disc revolutions rather than time. The fully flooded base oil result is also shown as a dashed line. At 20°C, see Fig. 4, film thickness drops rapidly in the first few minutes before stabilizing after typically 150 revolutions. At the slowest speed, the final grease film thickness (h_T) was greater than the corresponding base oil film thickness (h_{BO}), although at higher speeds this was reversed. The ratio h_T/h_{BO} generally decreases with speed, reflecting the increasing starvation of the inlet.

At 50°C, a similar rapid decrease in film thickness is seen up to 100 revolutions after which it starts to recover. Recovery occurs first at the slowest speed; this is to be expected as the resulting film thickness is a balance of oil release around the contact and time for this to occur. At the slower rolling speeds there is a longer time for the oil to reflow around the contact and this can compensate for poorer oil release.

The results at 80°C show very clearly the effects of improved grease supply to the contact at higher temperatures. At the slowest speed, the film tends to increase over the first five minutes, levelling off at 90–100 nm. Observation of the contact in these tests confirms that bulk grease flow into the inlet is maintained so that it is not severely starved. At the higher rolling speeds there is an initial decrease followed by recovery after ~80 revolutions for 0.05 ms^{-1} and after ~100 revolutions for 0.1 ms^{-1}.

A summary of representative film thickness results is shown in Table 2. Both h_T and h_R were measured at the end of the test, h_{BO} is the measured base oil value for a fully flooded contact and h_{EHD} the inferred grease hydrodynamic component. Generally, h_{EHD} decreases with increasing speed, showing the effects of increasing starvation. The residual films are often quite thick and generally reflect the degree of bulk grease flow into the contact.

h_{EHD} for the grease is plotted against rolling speed for each temperature in Figs. 7–9. The base oil results are also plotted as a solid line. At 20°C the grease EHD film component is less than the base oil and the ratio (h_{EHD}/h_{BO}) decreases with increasing speed. This reflects the increasing failure of the oil to replenish the contact as the available reflow time decreases at higher speeds. At higher temperatures h_{EHD} is usually greater than h_{BO}, suggesting that it is bulk grease rather than bled oil which is present in the inlet. The ratio

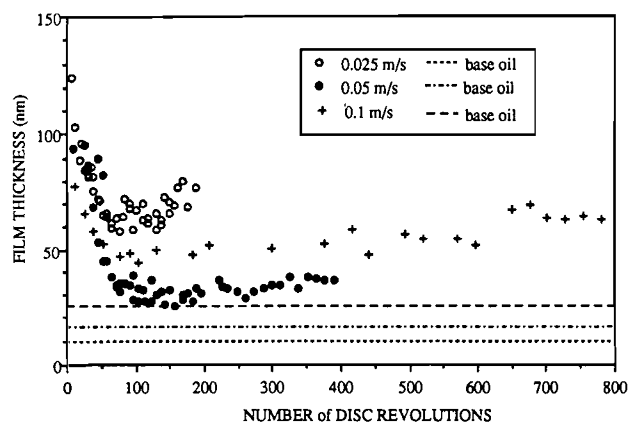

Fig. 5—Film thickness results for grease and base oil at 50°C.

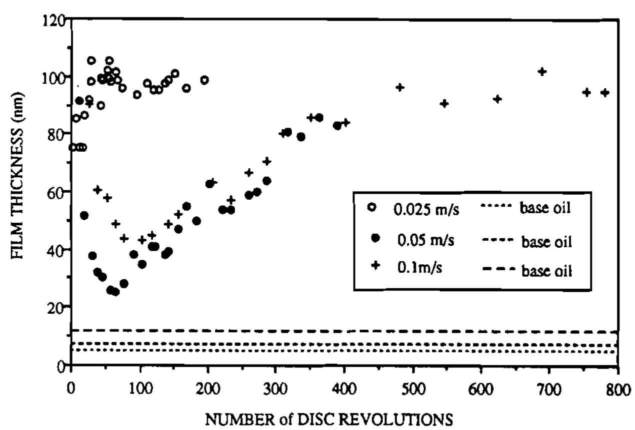

Fig. 6—Film thickness results for grease and base oil at 80°C.

TABLE 2—SUMMARY OF FILM THICKNESS RESULTS AFTER 30 MINUTES RUNNING

SPEED ms^{-1}	h_T nm	h_R nm	h_{EHD} nm	h_{BO} nm
(a) 20°C				
0.025	53	31	22	32
0.05	36	10	26	53
0.1	20	6	14	85
(b) 50°C				
0.025	77	57	20	10
0.05	33	8	25	16
0.1	54	33	19	25
(c) 80°C				
0.025	76	55	21	5
0.05	83	68	15	8
0.1	95	78	17	12

again decreases with speed as the system moves from grease to bled oil replenishment and is increasingly starved.

Meniscus Reformation Results

Reflow tests were run at the same temperature/speed levels as the EHD measurements. The motor was periodically stopped and the time taken for the oil to bleed from the grease back into the starved track was recorded. The reflow time plotted in the figures is the time taken for the oil to reach an arbitrary distance from the contact center. The distance chosen is different for each temperature so it is not possible to directly compare results across the temperature range. The inlet distance was chosen so that reflow time at the first measurement was within acceptable limits: neither too long (>240 seconds), as this would unacceptably prolong the test time, nor too short (<10 seconds) that any decrease would be difficult to record. For tests at 20°C the distance was 300 μm, 450 μm at 50°C and 500 μm at 80°C.

Generally, reflow time decreased with rolling time and bulk temperature. In the first few seconds of running the inlet is fully flooded so that measurements taken at the beginning are distorted by the uncleared grease, however, as the test proceeds, the inlet rapidly starves of bulk grease. The contact rapidly starves even at the lowest speeds at 20°C. However, reflow times decrease with increasing overrolling, suggesting that lubricant supply to the contact improves as the test proceeds. In Fig. 10(a) reflow time and grease film thickness are plotted on the same graph for the test condition 20°C/0.05 m/s. It is interesting to compare the time scales for the decreased reflow times and film thickness stabilization.

At higher temperatures it proved difficult to starve the inlet at the lowest speed. This observation is supported by the corresponding film thickness results where the EHD contribution is larger than predicted for the fully flooded base oil alone (see Figs. 8 and 9). This would suggest that bulk grease, albeit highly shear thinned, is being supplied to the contact.

In Figs. 10(b) and 10(c), reflow time and film thickness are compared for tests at 50° and 80°C (both at 0.05 ms^{-1}). In both cases, there is a good correspondence between the improvement in reflow time and film stabilization. For the test at 80°C, the inlet meniscus was reestablished 300–400 μm from the contact center after a few minutes rolling and this is indicated in Fig. 10(c). Under these conditions, it is grease rather than bled oil that is entrained into the inlet. This can be seen in the resulting increase in film thickness and, particularly, h_R.

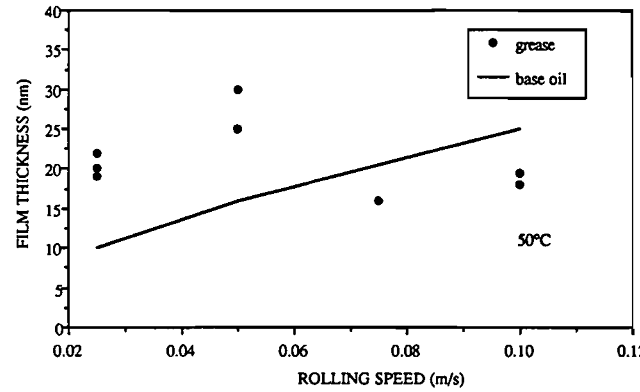

Fig. 8—Comparison of EHD film thickness for base oil and grease at 50°C.

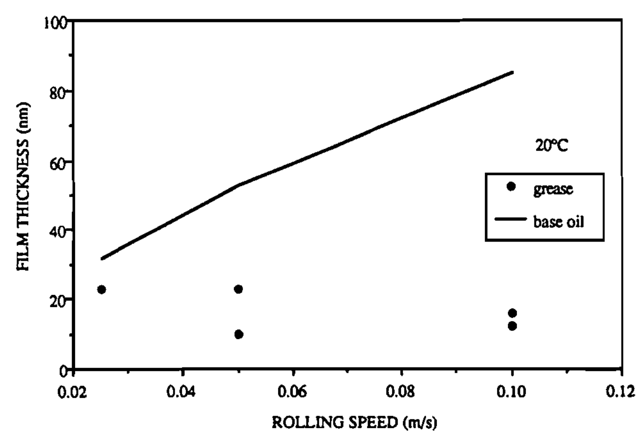

Fig. 7—Comparison of EHD film thickness for base oil and grease at 20°C.

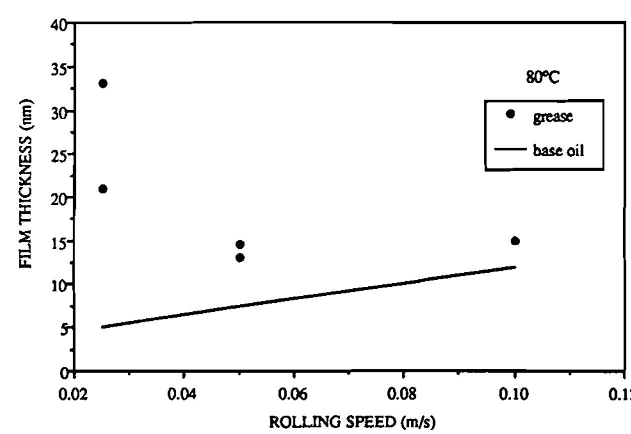

Fig. 9—Comparison of EHD film thickness for base oil and grease at 80°C.

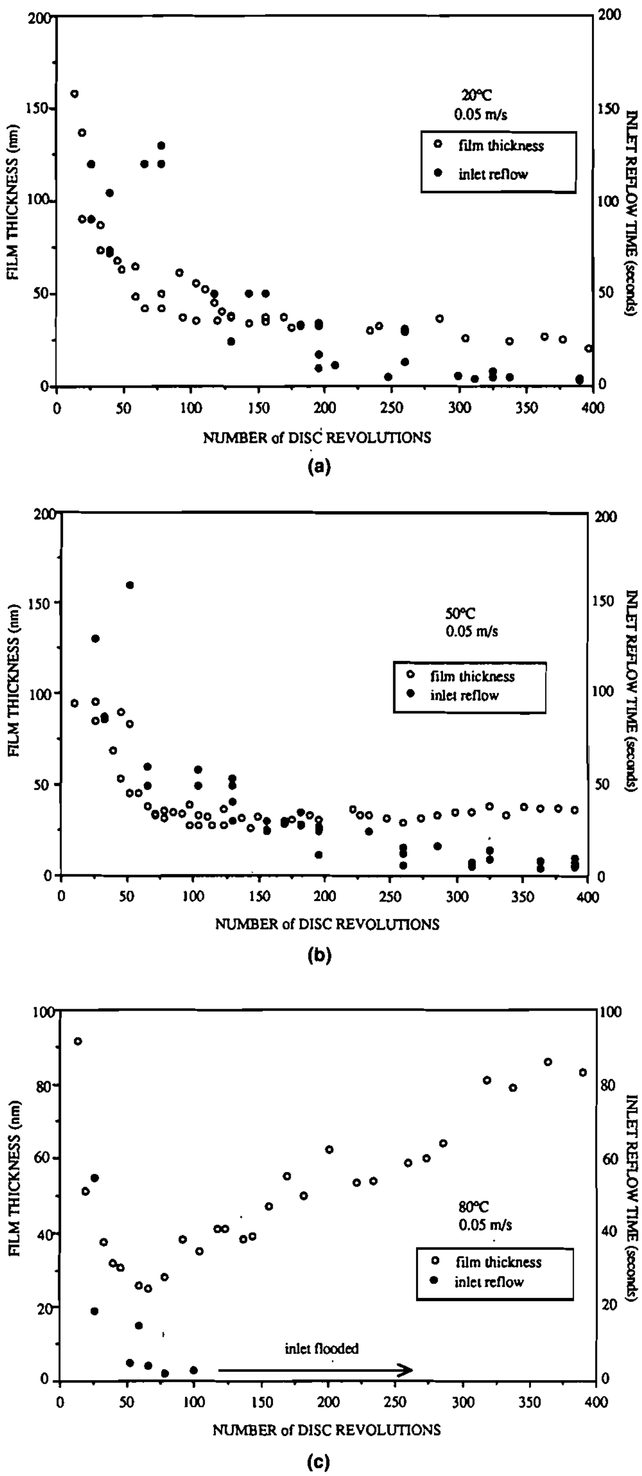

Fig. 10—Comparison of grease film thickness and reflow measurements for 0.05 m/s.
(a) 20°C
(b) 50°C
(c) 80°C

Discussion

The picture of greased lubrication that emerges from this work is essentially that of a flow balance and is shown in Fig. 11. The resulting net flow of lubricant either out of, or into, the contact determines the film thickness level. This model of grease behavior has obvious parallels with models for starved fluid film lubrication (*13*). In this work, two main film loss mechanisms were identified: inlet displacement of lubricant and cross or squeeze flow from the contact. The reflow mechanism in grease lubrication is more complex as it depends upon oil release from the grease and its availability close to the contact; hence, it is time dependent. Different regimes can be identified depending upon the dominant flow mechanisms, as Fig. 11 demonstrates. After the initial charge of grease, the bulk is rapidly pushed away from the track producing gross starvation of the inlet (I). Grease that has passed through the contact in the initial overrolling is shear degraded and deposited in the rolling track as thickener/oil glomerates and free base oil. This accords with the view expressed by Scarlett (*2*) of a highly worked grease layer deposited in the track. These structures are broken down and free oil is squeezed from the contact (II) with continued overrolling. Flow from the contact is inversely proportional to viscosity which correlates with observation from this work that film decay in the initial stages is faster at higher temperatures. At higher temperatures and longer rolling times, the grease structure to the side of the contact is progressively broken down facilitating oil release or bulk grease flow into the contact. It is possible for there now to be net flow back into the track (III).

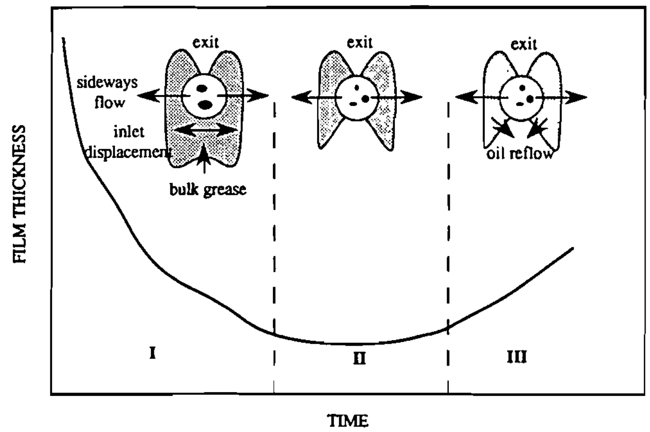

Fig. 11—Flow balance in grease lubrication.

Generally, inlet reformation time decreases with increasing temperature and increasing rotation time. This is due to the oil being released more readily by the overrolled (worked) grease and the decreasing viscosity of the base oil at higher temperatures. IR analysis of the oil that bleeds from the grease reservoir has shown it to be primarily base oil containing little thickener (*8*). This mechanism predominates when the bulk supply of grease to the inlet has failed and the contact is fully starved. The EHD component in this case is determined by the balance of oil lost from the contact due to squeeze flow and oil replenishment from the grease reservoir. However, if bulk flow into the inlet is maintained, then significant amounts of thickener flow into the contact which can increase the residual film. The grease EHD film thickness is now determined by the effective grease viscosity in the inlet and the degree of starvation.

CONCLUSIONS

From this and earlier work in this program, some general observations can be made about the nature of the films formed in rolling EHD contacts and the mechanisms that control film formation.

Nature of the Separating Films

There are two components:

1. A residual film which is speed independent and is primarily composed of degraded grease thickener deposited within the rolling track.
2. A hydrodynamic component, generated by the relative motion of the surfaces, due to oil both in the rolling track and supplied from the grease reservoir.

Film Thickness and Lubrication Mechanism

At low temperatures grease films are generally thinner than for the fully flooded base oil. This is due to the predominance of bulk grease starvation and the inability of the bled oil (due to high viscosity) to resupply the contact. At higher temperatures this situation is reversed with grease films considerably in excess of the base oil ones. This is attributed to the improved local supply of lubricant to the contact area at the higher temperature producing a partially flooded EHD film augmented by a boundary film of deposited thickener.

Grease lubrication film thicknesses in a rolling point contact are essentially controlled by the supply of lubricant, either grease or bled oil, to the contact. Two replenishment mechanisms have been observed in this work and both are controlled by the shear stability/temperature properties of the grease:

1. Bulk grease flow into the inlet due to the overrolling of the ball, forcing grease close to the track back into the inlet. This occurs during the first few minutes of rolling. It can also be reestablished for high temperature/low speed conditions once the grease close to the track has been worked sufficiently.
2. Local reflow of oil bleeding from the grease close to the track. Replenishment is determined by the capacity of the grease to release the oil and the way this is affected by operating conditions (degree of working and bulk temperature). This controls the establishment of the primary reservoirs noted by Åström (*6*). Local replenishment into the inlet is probably dominated by the bled oil properties, particularly the viscosity. However, the effect of overrolling or working on the grease close to the contact is important as is this response which controls oil replenishment to the contact.

Both Mechanisms 1 and 2 can starve as the rolling speed increases and the effective time for reflow decreases, resulting in lower film thicknesses.

Such findings have significant implications for the assessment of the lubrication requirements for bearings which are often based on the base oil viscosity. These results have shown that such a simple assessment of lubrication level is not sufficient, even for a model rolling contact. This paper is a study of grease lubrication in a model slow speed contact; whether these ideas and observations can be applied to the far more complex problem of bearing lubrication is, at present, a matter of conjecture.

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